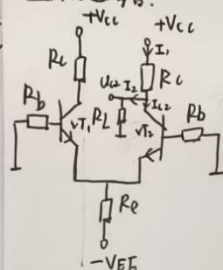


四种差分放大电路详细讲解请参考网址

https://blog.csdn.net/weixin_43348794/article/details/150108099

1. 直流通路:



VT1管:

$$0 - (-V_{EE}) = R_b \cdot I_{BQ} + U_{BEQ} = 2 I_{EQ} R_E \quad \text{①}$$

$$I_E = (1 + \beta) I_B \quad \text{②}$$

得 $I_B = \quad I_E =$

VT2管:

$$I_1 = I_2 + I_{L2}$$

$$I_{L2} = \beta I_{BQ}$$

$$V_{CC} - U_{C2}' = I_1 R_L$$

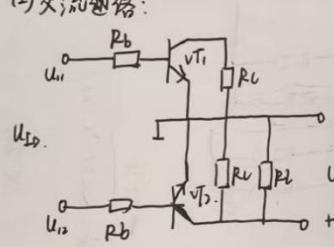
$$U_{C2}' = I_2 R_L$$

得 $U_{C2}' = \quad I_{L2} =$

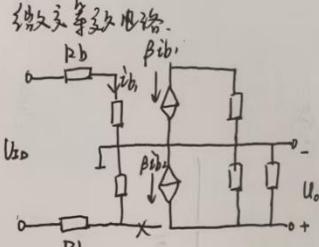
$$U_{C2}' - (-V_{EE}) = U_{CE2} + 2 I_E R_E$$

得 $U_{CE2} =$

(2) 交流通路:

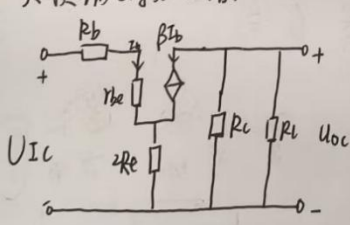


微变等效电路:



$$A_{ud} = \frac{u_{od}}{u_{io}} = \frac{\beta (R_L \parallel R_L)}{2(R_b + r_{be})} =$$

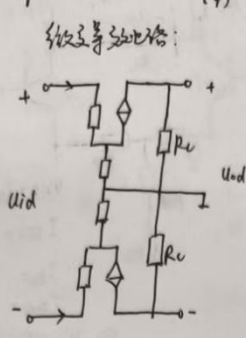
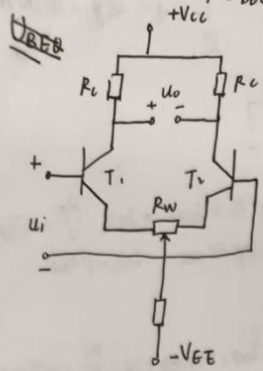
共模微变等效电路:



$$A_{uc} = \frac{u_{oc}}{u_{ic}} = - \frac{\beta (R_L \parallel R_L)}{R_b + r_{be} + 2(1 + \beta) R_E}$$

(3) 略.

$I_{BQ} = 100 \mu A, U_{BEQ} = 0.7 V.$



(7) $U_{BEQ} + I_{EQ} \cdot \frac{R_W}{2} + 2I_{EQ} \cdot R_E = V_{EE}$

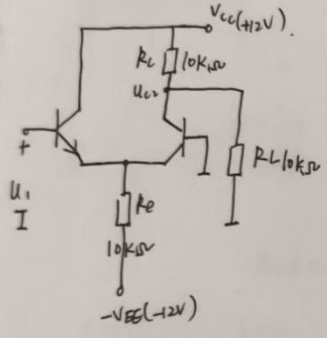
$I_{EQ} = \frac{V_{EE} - U_{BEQ}}{\frac{R_W}{2} + 2R_E} \approx 0.517 mA$

$r_{be} = r_{bb'} + (1+\beta) \frac{26mV}{I_{EQ}} \approx 5.18 K\Omega$

$A_d = - \frac{\beta R_C}{r_{be} + (1+\beta) R_W/2} \approx -99$

$R_i = 2r_{be} + (1+\beta) R_W$

3. $\beta = 60, r_{be1} = r_{be2} = 2 K\Omega$



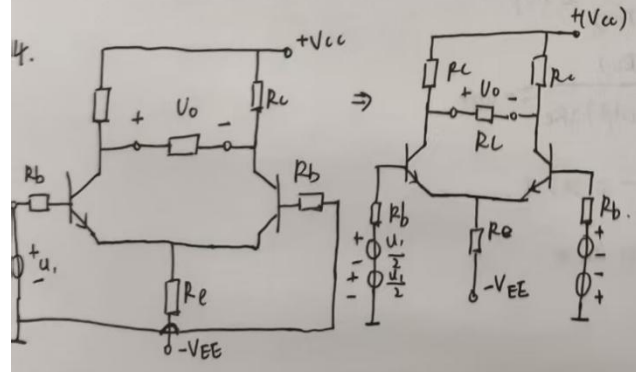
$A_{ud} = \frac{\beta(R_C // R_L)}{2r_{be}} = 75$

图略:

$A_{uc} = - \frac{\beta(R_C // R_L)}{r_{be} + (1+\beta) R_E} \approx -1.5$

$K_{CMR} = \frac{A_{ud}}{A_{uc}}$

4.



等效为双端输入.

$A_{ud} = - \frac{\beta(R_C // \frac{R_L}{2})}{R_B + r_{be}}$

$A_{uc} = 0$

$K_{CMR} = \infty$

5.

$A_u = -100$. 饱和电压: $U_{CES} = 1V$.
 (1) 不失真输出电压有效值 $U_{om} = \frac{V_{CC} - U_{CES}}{\sqrt{2}} \approx 7.78V$
 输入电压最大有效值: $U_{im} = \frac{U_{om}}{|A_u|} \approx 77.8mV$.
 (2) $U_i = 10mV$, $U_o = 100 \cdot U_i = 1V$.
 (注: 此题意义不大, 请掌握PPT第37页的倍频电路, 以及电路设计).
 静态: $U_{B1B2} = U_{P21} + U_{D22}$
 动态: $u_{b1} \approx u_{b2} \approx u_i$

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